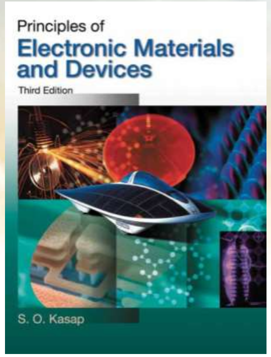
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》
Solid-State Electronics



Ch2-2
Electrical & Thermal Conduction in Solids

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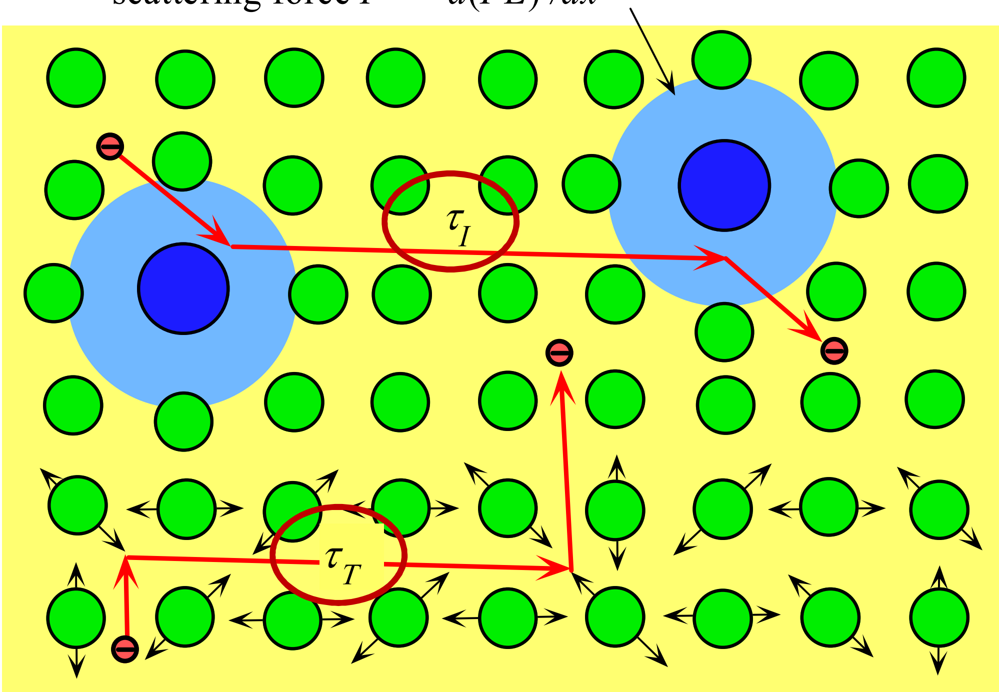
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

2.3.1 Matthiessen's rule for metallic alloy
金属合金固溶体的马希森定则



$$F = -\frac{d(PE)}{dx}$$

Two different types of scattering processes involving scattering from impurities alone and thermal vibrations alone.

As long as the impurity atom results in a local distortion of the crystal lattice, it will be effective in scattering.

$$\frac{1}{\tau} = \frac{1}{\tau_T} + \frac{1}{\tau_I}$$

$\frac{1}{\tau}$: the net probability of scattering

$\frac{1}{\tau_T}$: the probability of scattering from lattice vibrations

$\frac{1}{\tau_I}$: the probability of scattering from impurities

$$\frac{e\tau}{m_e} = \mu_d$$ dirft mobility 迁移率

$$\begin{cases} \frac{e\tau_T}{m_e} = \mu_L & \text{Lattice-scattering-limited drift mobility} \\ \frac{e\tau_I}{m_e} = \mu_I & \text{Impurity-scattering-limited drift mobility} \end{cases}$$

$$\frac{1}{\mu_d} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$$

Resistivity $\rho = \dfrac{1}{en\mu_d} = \dfrac{1}{en\mu_L} + \dfrac{1}{en\mu_I}$

definition $\rho_T = \dfrac{1}{en\mu_L}$ $\rho_I = \dfrac{1}{en\mu_I}$

$$\Rightarrow \quad \rho = \rho_T + \rho_I$$

$\rho = \rho_T + \rho_R$ Matthiessen's rule （马希森定则）

ρ_R: the residual resistivity, due to the scattering of electrons by impurities, dislocations, interstitial atoms, vacancies, grain boundaries, etc.

$$\rho \approx AT + B$$

The fractional change in the resistivity per unit temperature increase at the reference temperature T_0

在参考温度温度附近单位温度变化时电阻率变化的百分比

$$a_0 = \frac{1}{\rho_0}\left[\frac{\delta\rho}{\delta T}\right]_{T=T_0}$$

Temperate coefficient of resistivity (TCR)

电阻率温度系数

Table 2.1 Resistivity, thermal coefficient of resistivity α_0 at 273 K (0 °C) for various metals. The resistivity index n in $\rho \propto T^n$ for some of the metals is also shown.

Metal	ρ_0 (nΩ m)	$\alpha_0\left(\frac{1}{K}\right)$	n	Comment
Aluminum, Al	25.0	$\frac{1}{233}$	1.20	
Antimony, Sb	38	$\frac{1}{196}$	1.40	
Copper, Cu	15.7	$\frac{1}{232}$	1.15	
Gold, Au	22.8	$\frac{1}{251}$	1.11	
Indium, In	78.0	$\frac{1}{196}$	1.40	
Platinum, Pt	98	$\frac{1}{255}$	0.94	
Silver, Ag	14.6	$\frac{1}{244}$	1.11	
Tantalum, Ta	117	$\frac{1}{294}$	0.93	
Tin, Sn	110	$\frac{1}{217}$	1.11	
Tungsten, W	50	$\frac{1}{220}$	1.20	
Iron, Fe	84.0	$\frac{1}{152}$	1.80	Magnetic metal; $273 < T < 1043$ K
Nickel, Ni	59.0	$\frac{1}{125}$	1.72	Magnetic metal; $273 < T < 627$ K

SOURCE: Data were extracted and combined from several sources. Typical values.

for metal

n~1

$$\rho = \rho_0\left[1 + a_0\left(T - T_0\right)\right]$$

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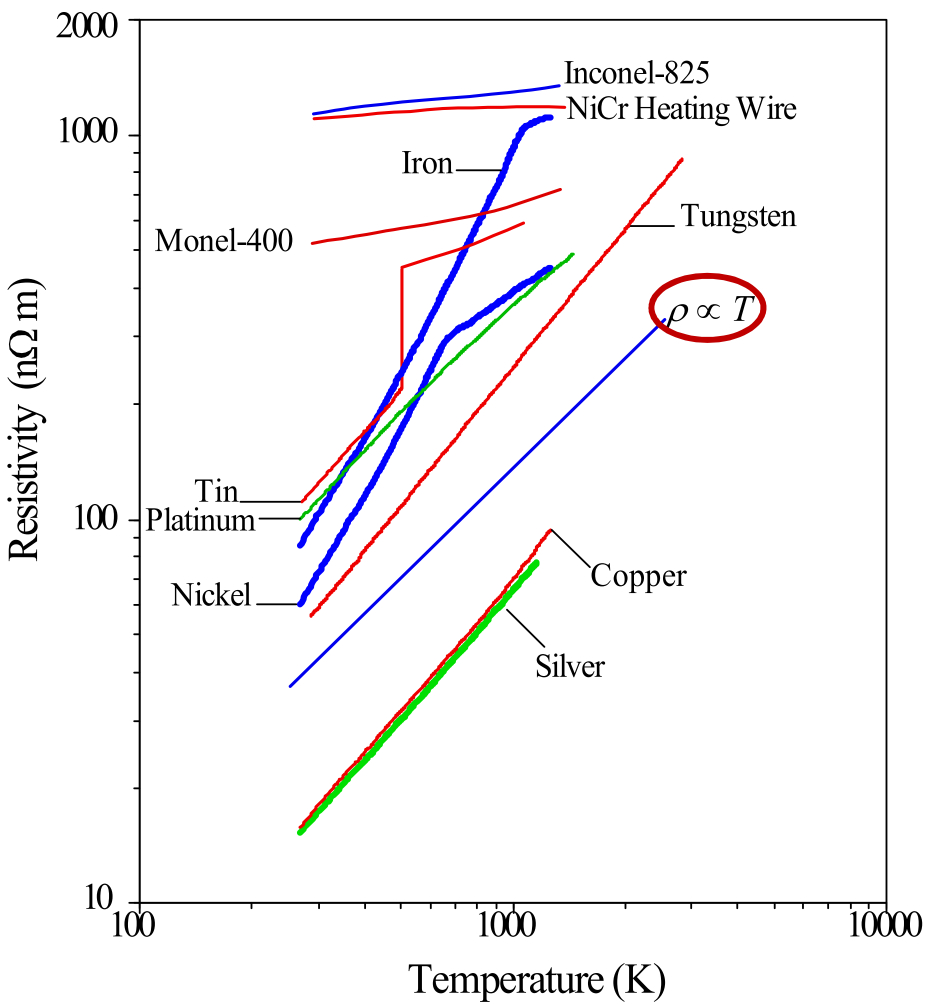
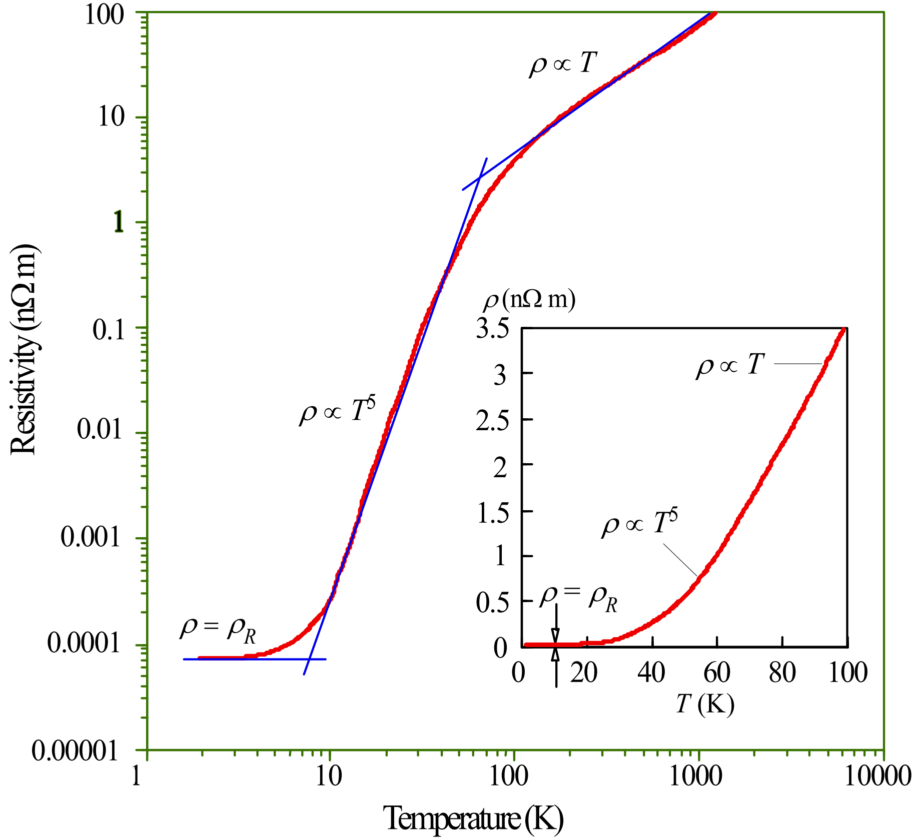


$$\rho = \rho_0 \left[\frac{T}{T_0} \right]^n$$

The resistivity of various metals as a function of temperature above 0 °C. Tin melts at 505 K whereas nickel and iron go through a magnetic to non-magnetic (Curie) transformations at about 627 K and 1043 K respectively. The theoretical behavior ($\rho \sim T$) is shown for reference.



$$\rho \approx AT + B$$

$$\rho \propto T^5$$

$$\rho = DT^5 + \rho_R$$

The resistivity of copper from lowest to highest temperatures (near melting temperature, 1358 K) on a log-log plot. Above about 100 K, $\rho \propto T$, whereas at low temperatures, $\rho \propto T^5$ and at the lowest temperatures ρ approaches the residual resistivity ρ_R. The inset shows the ρ vs. T behavior below 100 K on a linear plot (ρ_R is too small on this scale).

Metal's Resistivity and Purity

The total resistance of a metal includes the basic resistance (residual resistance) of the metal and the solute (impurity) resistance.

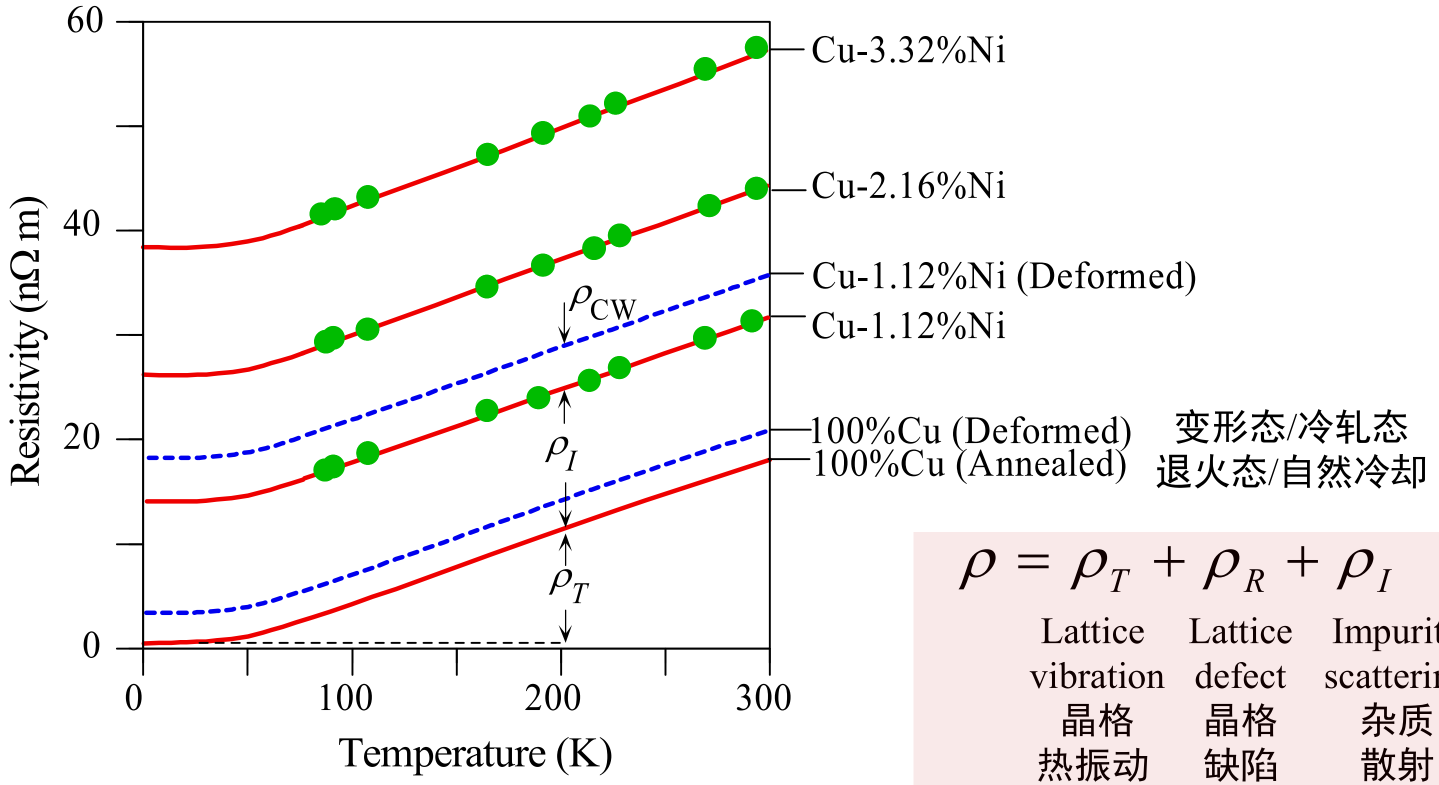
Matthiessen's rule $\rho = AT + B = \rho_0 + \rho'(T)$

- ρ_0: Resistivity associated with impurities
- $\rho'(T)$: Temperature dependent resistivity

Conclusion

1. At low temperature, ρ_0 is s a main factor: The resistivity measured at very low temperatures (4.2 K); referred to as the metal residual resistivity; it reflects the purity of the metal.
2. At high temperature, $\rho'(T)$ is a main factor

Explain the typical resistivity versus temperature behavior of annealed and cold-worked (CW) copper containing various amount of Ni.



$$\rho = \rho_T + \rho_R + \rho_I$$

ρ_T	ρ_R	ρ_I
Lattice vibration 晶格热振动	Lattice defect 晶格缺陷	Impurity scattering 杂质散射

Typical temperature dependence of the resistivity of annealed and cold worked (deformed) copper containing various amount of Ni in atomic percentage (data adapted from J.O. Linde, *Ann. Pkysik*, 5, 219 (1932)).

EXAMPLE 2.8 **TEMPERATURE COEFFICIENT OF RESISTIVITY α AND RESISTIVITY INDEX n** If α_0 is the temperature coefficient of resistivity (TCR) at temperature T_0 and the resistivity obeys the equation

$$\rho = \rho_0 \left[\frac{T}{T_0}\right]^n$$

show that

$$\alpha_0 = \frac{n}{T_0}\left[\frac{T}{T_0}\right]^{n-1}$$

What is your conclusion?

Experiments indicate that $n = 1.2$ for W. What is its α_0 at 20 °C? Given that, experimentally, $\alpha_0 = 0.00393$ for Cu at 20 °C, what is n?

SOLUTION

Since the resistivity obeys $\rho = \rho_0(T/T_0)^n$, we substitute this equation into the definition of TCR,

$$\boxed{\alpha_0 = \frac{1}{\rho_0}\left[\frac{d\rho}{dT}\right]} = \frac{n}{T_0}\left[\frac{T}{T_0}\right]^{n-1}$$

It is clear that, in general, α_0 depends on the temperature T, as well as on the reference temperature T_0. The TCR is only independent of T when $n = 1$.

At $T = T_0$, we have

$$\frac{\alpha_0 T_0}{n} = 1 \qquad \text{or} \qquad n = \alpha_0 T_0$$

For W, $n = 1.2$, so at $T = T_0 = 293$ K, we have $\alpha_{293\ \mathrm{K}} = 0.0041$, which agrees reasonably well with $\alpha_{293\ \mathrm{K}} = 0.0045$, frequently found in data books.

For Cu, $\alpha_{293\ \mathrm{K}} = 0.00393$, so that $n = 1.15$, which agrees with the experimental value of n.

TEMPERATURE OF THE FILAMENT OF A LIGHT BULB

EXAMPLE 2.10

a. Consider a 40 W, 120 V incandescent light bulb. The tungsten filament is 0.381 m long and has a diameter of 33 μm. Its resistivity at room temperature is 5.51×10^{-8} Ω m. Given that the resistivity of the tungsten filament varies at $T^{1.2}$, estimate the temperature of the bulb when it is operated at the rated voltage, that is, when it is lit directly from a power outlet, as shown schematically in Figure 2.10. Note that the bulb dissipates 40 W at 120 V.

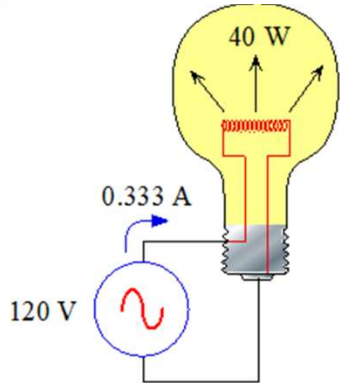
b. Assume that the electrical power dissipated in the tungsten wire is radiated from the surface of the filament. The radiated electromagnetic power at the absolute temperature T can be described by **Stefan's law,** as follows:

$$P_{\text{radiated}} = \epsilon \sigma_S A(T^4 - T_0^4)$$

where σ_S is Stefan's constant (5.67×10^{-8} W m^{-2} K^{-4}), ϵ is the emissivity of the surface (0.35 for tungsten), A is the surface area of the tungsten filament, and T_0 is the room temperature (293 K). For $T^4 \gg T_0^4$, the equation becomes

$$P_{\text{radiated}} = \epsilon \sigma_S A T^4$$

Assuming that all the electrical power is radiated as electromagnetic waves from the surface, estimate the temperature of the filament and compare it with your answer in part (a).



SOLUTION

a. When the bulb is operating at 120 V, it is dissipating 40 W, which means that the current is

$$I = \frac{P}{V} = \frac{40\ \text{W}}{120\ \text{V}} = 0.333\ \text{A}$$

The resistance of the filament at the operating temperature T must be

$$R = \frac{V}{I} = \frac{120}{0.333} = 360\ \Omega$$

Since $R = \rho L/A$, the resistivity of tungsten at the operating temperature T must be

$$\rho(T) = \frac{R(\pi D^2/4)}{L} = \frac{360\ \Omega\,\pi(33 \times 10^{-6}\ \text{m})^2}{4(0.381\ \text{m})} = 8.08 \times 10^{-7}\ \Omega\,\text{m}$$

But, $\rho(T) = \rho_0(T/T_0)^{1.2}$, so that

$$T = T_0\left(\frac{80.8 \times 10^{-8}}{5.51 \times 10^{-8}}\right)^{1/1.2}$$

$$= 2746\ \text{K} \qquad \text{or} \qquad 2473\ ^\circ\text{C}$$ (melting temperature of W is about 3680, K)

b. To calculate T from the radiation law, we note that $T = [P_{\text{radiated}}/\epsilon\sigma_S A]^{1/4}$.
The surface area is

$$A = L(\pi D) = (0.381)(\pi 33 \times 10^{-6}) = 3.95 \times 10^{-5}\ \text{m}^2$$

Then,

$$T = \left[\frac{P_{\text{radiated}}}{\epsilon\sigma_S A}\right]^{1/4} = \left[\frac{40\ \text{W}}{(0.35)(5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4})(3.95 \times 10^{-5}\ \text{m}^2)}\right]^{1/4}$$

$$= [5.103 \times 10^{13}]^{1/4} = 2673\ \text{K} \qquad \text{or} \qquad 2400\ ^\circ\text{C}$$

The difference between the two methods is less than 3 percent.

2.3.2 Solid solutions and Nordheim's rule 固溶体与诺德海姆定则

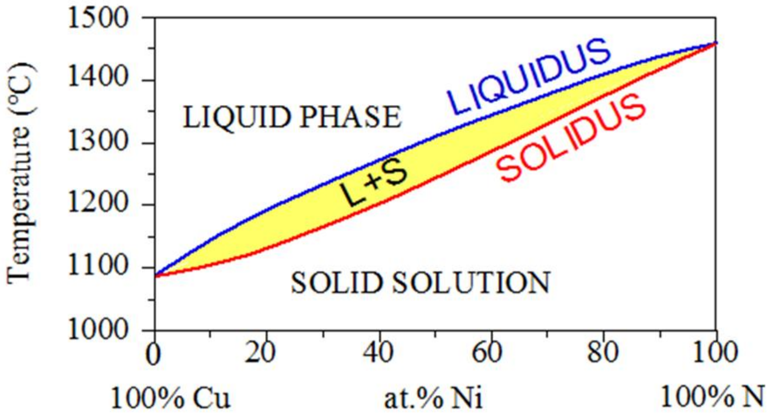
Change in resistivity when forming a solid solution

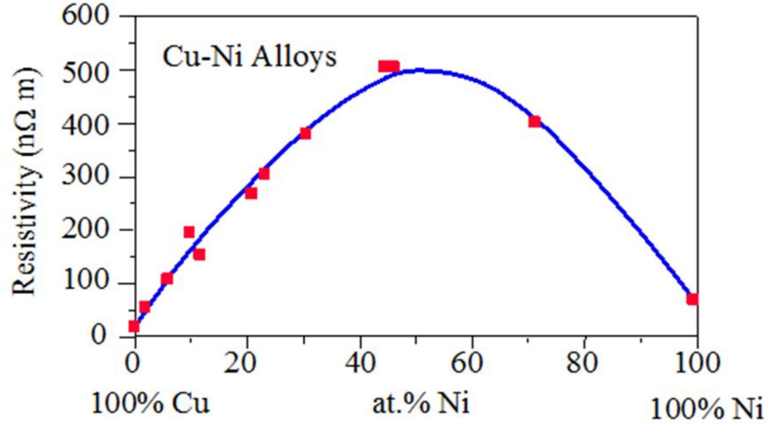
Table 2.2 The effect of alloying on the resistivity

$$\rho = \rho_T + \rho_I$$

Material	Resistivity at 20 °C (nΩ m)	α at 20 °C (1/K)
Nickel	69	0.006
Chrome	129	0.003
Nichrome	1120	0.0003

- When a solid solution is formed, the electrical conductivity of the alloy is lowered. This is true even when a highly conductive solute metal is dissolved in a conductive metal solvent.
- This is because when the solute atoms are dissolved in the solvent crystal lattice, the lattice of the solvent is distorted, disrupting the periodicity of the lattice potential field, thereby increasing the probability of electron scattering and increasing the resistivity. However, lattice distortion is not the only factor in the change of resistivity. The electrical properties of solid solution still depend on the chemical interactions (bands, electron cloud distribution, etc.) of solid solution components.





(a) Phase diagram of the Cu-Ni alloy system. Above the liquidus line only the liquid phase exists. In the $L + S$ region, the liquid (L) and solid (S) phases coexist whereas below the solidus line, only the solid phase (a solid solution) exists. (b) The resistivity of the Cu-Ni alloy as a function of Ni content (at.%) at room temperature. [Data extracted from Metals Handbook-10th Edition, Vols 2 and 3, ASM, Metals Park, Ohio, 1991 and Constitution of Binary Alloys, M. Hansen and K. Anderko, McGraw-Hill, New York, 1958]

The maximum resistivity in a binary alloy is at 50% atomic concentration, which may be several times higher than the component resistivity.

The solid solution composed of ferromagnetism and strong paramagnetic metal is abnormal, and its resistivity is generally not at 50%.

$$\rho_I = CX\left(1 - X\right) \qquad \text{Nordheim's rule}$$

Table 2.3 Nordheim coefficient C (at 20 °C) for dilute alloys obtained from $\rho_I = CX$ and $X < 1$ at.%*

Solute in Solvent (element in matrix)	C (nΩ m)	Maximum Solubility at 25 °C (at.%)
Au in Cu matrix	5500	100
Mn in Cu matrix	2900	24
Ni in Cu matrix	1200	100
Sn in Cu matrix	2900	0.6
Zn in Cu matrix	300	30
Cu in Au matrix	450	100
Mn in Au matrix	2410	25
Ni in Au matrix	790	100
Sn in Au matrix	3360	5
Zn in Au matrix	950	15

*NOTE: For many isomorphous alloys C may be different at higher concentrations; that is, it may depend on the composition of the alloy.

SOURCES: D.G. Fink and D. Christiansen, eds., *Electronics Engineers' Handbook*, 2nd ed., New York, McGraw-Hill, 1982. J. K. Stanley, *Electrical and Magnetic Properties of Metals*, Metals Park, OH, American Society for Metals, 1963. Solubility data from M. Hansen and K. Anderko, *Constitution of Binary Alloys*, 2nd ed., New York, McGraw-Hill, 1985.

$$\rho = \rho_{matrix} + CX(1-X)$$
$$= \underbrace{\rho_T + \rho_R}_{\text{Matthiessen}} + \underbrace{CX(1-X)}_{\text{Nordheim}}$$

ρ_{matrix} : the resistivity of the matrix due to scattering from thermal vibrations and from other defects.

质量分数

EXAMPLE 2.11 **NORDHEIM'S RULE** The alloy 90 wt.% Au–10 wt.% Cu is sometimes used in low-voltage dc electrical contacts, because pure gold is mechanically soft and the addition of copper increases the hardness of the metal without sacrificing the corrosion resistance. Predict the resistivity of the alloy and compare it with the experimental value of 108 nΩ m.

SOLUTION

We apply Equation 2.22, $\rho(X) = \rho_{Au} + CX(1 - X)$ but with 10 wt.% Cu converted to the atomic fraction for X. If w is the weight fraction of Cu, $w = 0.1$, and if M_{Au} and M_{Cu} are the atomic masses of Au and Cu, then the atomic fraction X of Cu is given by (see Example 1.2),

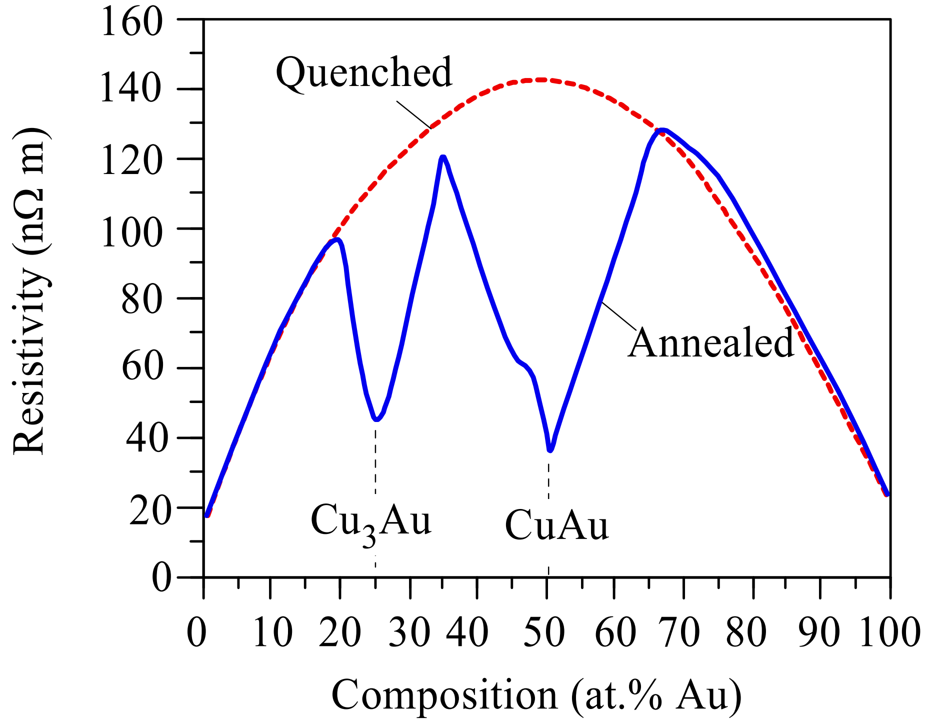
原子分数 $$X = \frac{w/M_{Cu}}{w/M_{Cu} + (1 - w)/M_{Au}} = \frac{0.1/63.55}{(0.1/63.55) + (0.90/197)} = 0.256$$

Given that $\rho_{Au} = 22.8$ nΩ m and $C = 450$ nΩ m,

$$\rho = \rho_{Au} + CX(1 - X) = (22.8 \text{ n}\Omega\text{ m}) + (450 \text{ n}\Omega\text{ m})(0.256)(1 - 0.256)$$

$$= 108.5 \text{ n}\Omega\text{ m}$$

This value is only 0.5% different from the experimental value.



Ordered crystalline structure
有序晶体结构

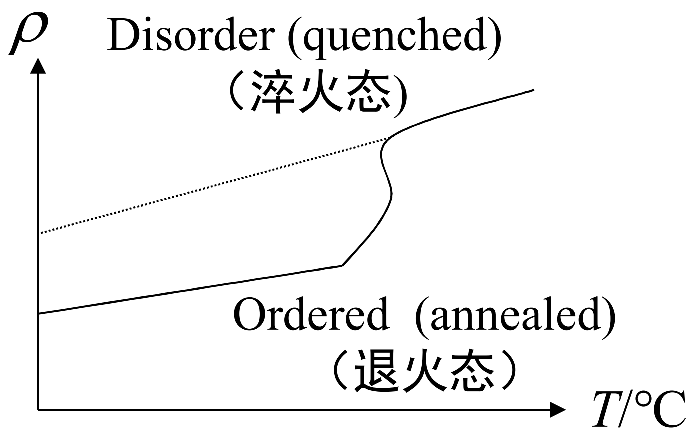
Electrical resistivity vs. composition at room temperature in Cu-Au alloys. The quenched sample (dashed curve) is obtained by quenching the liquid and has the Cu and Au atoms randomly mixed. The resistivity obeys the Nordheim rule. On the other hand, when the quenched sample is annealed or the liquid slowly cooled (solid curve), certain compositions (Cu_3Au and CuAu) result in an ordered crystalline structure in which Cu and Au atoms are positioned in an ordered fashion in the crystal and the scattering effect is reduced.

Resistivity of Ordered Alloy

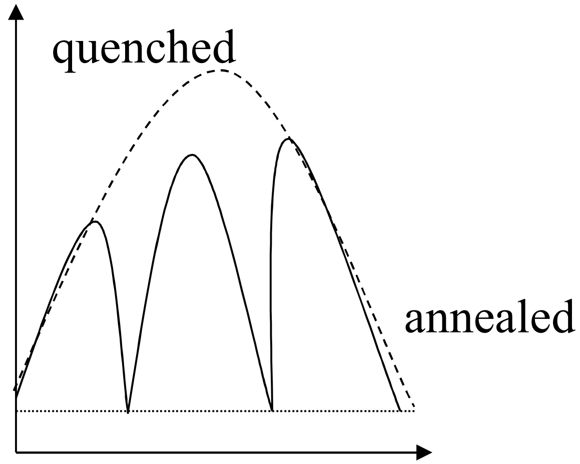
For ordered solid solution, its alloy component's chemical action is strengthened, so

- The binding of electrons is stronger than in the disordered state, which reduces the number of conductive electrons and increases the residual resistance of the alloy.
- However, the crystal potential field is more symmetrical when ordered, which greatly reduces the electron scattering probability, and thus the residual resistance of the ordered alloy decreases.

Generally, in the above two opposite effects, the effect of the second factor predominates, so when the alloy is ordered, the resistivity is lowered.



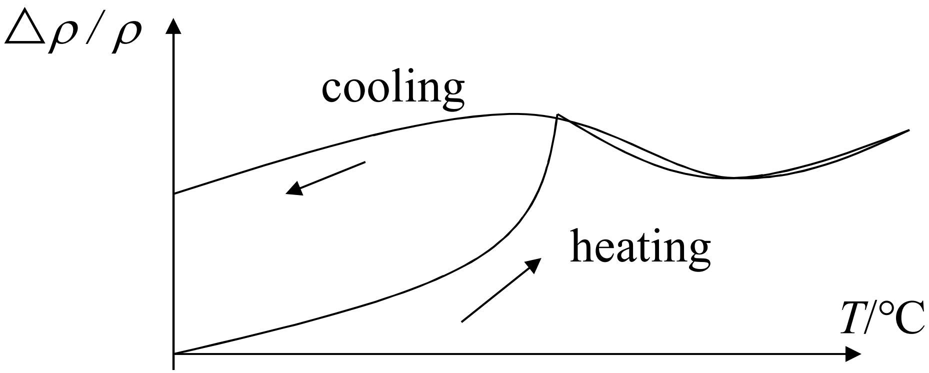
Ordering affects resistivity of Cu_3Au alloy



Resistivity of Cu-Au alloy

Resistivity of Uneven Solid Solution

- In alloys containing transition metals, such as nickel-chromium, nickel-copper-zinc, iron-chromium-aluminum, iron-nickel-molybdenum, silver-manganese, etc., according to X-ray and electron microscopy analysis, they are considered to be single phase. However, during the cooling process, the resistance of the alloy is abnormally increased (other physical properties, such as thermal expansion effect, specific heat capacity, elasticity, internal friction, etc.) also have significant changes. For the cold-worked（冷加工）, the alloy resistivity was found to be significantly reduced. Thomas first discovered that this state of the structure is called the K state.
- The cold-worked treatment greatly promotes the destruction of the uneven structure of the solid solution and obtains a common disordered solid solution, so the electrical resistivity is remarkably lowered.



Resistivity change of thermal treatment of 80Ni20Cr alloy

Metal Resistivity and Pressure

Resistivity and pressure： $\rho(P)=\rho_0(1+\alpha P)$

- ρ_0： resistivity at vacuum
- P： pressure
- α： The resistance pressure coefficient, generally -10^{-5}~-10^{-6}, can be a positive number

1. When the resistance coefficient of the resistor is negative, the resistivity decreases with increasing pressure, which is called **normal metal** （正常金属）, such as gold, silver, copper, iron, and so on.
2. When the resistance coefficient of the resistor is positive, the resistivity increases with the increase of the pressure, which is called an **abnormal metal** （反常金属）, such as an alkali metal （碱金属）, rare earth metal, calcium strontium etc.

Metal Resistivity and Pressure

- In hydrostatic compression (up to 1.2 GPa), the resistance of most metals drops. This is because under the condition of large hydrostatic pressure, the metal atom spacing is reduced, and the internal defect morphology, electronic structure, Fermi energy and band structure will change, which obviously affects the conductivity of the metal.
- The resistivity of a metal under hydrostatic pressure can be calculated by the following formula
- $\rho(P) = \rho_0(1+aP)$; where ρ_0 represents the resistivity under vacuum conditions, P represents pressure, a is the pressure coefficient (negative value $10^{-5} \sim 10^{-6}$)

- A big pressure can cause many substances to change from semiconductors and insulators to conductors and even superconductors.